Exact Observation Depth for Integer-Height Graph Populations

Connected supports, delayed distinctions, and sharp reconstruction

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Abstract

For a finite simple graph, we count nonnegative integer vertex heights jointly by their sum and a pinned quadratic edge energy. We prove a connected-support expansion for the formal logarithm: the coefficients through total height R depend only on connected induced vertex sets of size at most R, decorated with their ambient degrees. This gives an exact bounded-degree compiler whose graph-size cost is linear at fixed R. The expansion identifies the first coefficient at which equal-order regular graphs of girth at least g can differ: it is the difference of their g-cycle counts times an explicit polynomial. Our central reconstruction theorem determines every n-vertex simple 2-regular graph from the pinned series through volume floor(n/2), at every fixed real energy weight strictly between zero and one. The budget is sharp for every n at least six. In particular, a cycle of length 2g and two disjoint g-cycles agree below volume g and have a nonzero explicit discrepancy at g. Their dominating-terminal blocks remain indistinguishable below that volume under every fixed closing context. We supply complete formal-series proofs, independent finite enumeration receipts, cycle reconstruction for all 57 simple 2-regular graphs on at most fourteen vertices, and retained bounded-degree timing comparisons. The connected-cluster principle and low-order graph-coefficient methods have established antecedents; the claim here is the stated height-population specialization and its exact reconstruction and observation-depth theorems.

1 Population, budget and contribution

Let H=(V,E) be a finite simple loopless graph. Heights are functions h:V to the nonnegative integers. They remain attached to the original labeled vertices; no quotient by graph automorphisms is taken. Define

$$J(h)=\sum_{v\in V} h_v, \qquad E_H(h)=\sum_{uv\in E}(h_u-h_v)^2,$$

and, for an integer $\lambda \geq 0$,

$$F_\lambda(H;x,q)=\sum_h x^{J(h)} q^{E_H(h)+\lambda\sum_v h_v^2}. \tag{1}$$

The coefficient of x^j is a polynomial in q with nonnegative integer coefficients. It counts a finite population of $\binom{j+n-1}{n-1}$ profiles when $n=|V|>0$. All identities below

are initially identities in $\mathbb{Q}[q][[x]]$; evaluation at $q_0 \in (0,1)$ is coefficientwise. No infinite analytic expansion or convergence assertion is needed. We write $A_H = F_2(H)$ and $L_H = F_0(H)$.

A budget R means that coefficients of x through degree R are available. It bounds total integer height, rather than the number of occupied vertices alone or the maximum height. A profile at volume R occupies at most R vertices. The logarithm strengthens this elementary observation: only connected occupied footprints need to be compiled independently. We exploit that identity to establish a sharp observation threshold for an infinite graph class.

The broad computational idea is established. Patel and Regts [PR2017, Section 3] compute low-order graph-polynomial information from connected induced subgraph counts, including colored variants. Their additivity criterion explains the vanishing of disconnected terms. Their later induced-count algorithm [PR2019] supplies further bounded-degree context. Our ambient-degree labels and unbounded-height formal series require the specific specialization proved below; we do not claim a new connected-cluster principle. Bauerschmidt, Park and Rodriguez [BPR2024] study the integer-valued Gaussian model on lattices and its high-temperature scaling, a different question from finite-volume exact graph reconstruction. Chertkov and Chernyak [CC2006] express partition functions through belief-propagation loop corrections. The present paper uses a direct formal logarithm and does not claim to originate loop expansions or to provide a new belief-propagation approximation.

The principal result is Theorem 6: the exact and sharp budget $\lfloor n/2 \rfloor$ for simple 2-regular interiors of known order n. The preceding local identities provide its proof and an implementation. General completeness of the single fixed-pinning series A_H is not asserted.

2 Exact connected-support expansion

Expand the exponent in (1) as

$$\sum_v w_v h_v^2 - 2 \sum_{uv \in E} h_u h_v, \qquad w_v = d_H(v) + \lambda. \tag{2}$$

For $T \subseteq V$, let Z_T be (1) with heights outside T forced to zero. It is essential that (2) retains the ambient degree $d_H(v)$, rather than replacing it by the degree in $H[T]$: an edge to a vertex forced to height zero still contributes energy. Set $Z_\emptyset = 1$, and define

$$\Phi_S = \sum_{T \subseteq S} (-1)^{|S|-|T|} \log Z_T. \tag{3}$$

The logarithm exists formally because each series has constant coefficient one.

Theorem 1 (connected-support locality; SNG-F55). For every nonempty S, Φ_S vanishes when $H[S]$ is disconnected, has no term below degree $|S|$, and depends only on $H[S]$ decorated by w_v. Moreover,

$$\log F_\lambda(H) = \sum_{\substack{\emptyset \neq S \subseteq V \\ H[S] \text{ connected}}} \Phi_S. \tag{4}$$

Thus the coefficients through degree R are determined by the multiplicities of decorated connected induced subgraphs with at most R vertices.

Proof. Give each vertex its own variable y_v, replacing $x^{J(h)}$ by $\prod_v y_v^{h_v}$, and call the resulting series F. Forcing heights outside T to zero is the substitution $y_v = 0$ outside T. The substitution commutes with formal logarithms. A monomial of $\log F$ supported on U therefore survives in $\log Z_T$ exactly when $U \subseteq T$. Its multiplicity in (3) is

$$\sum_{U \subseteq T \subseteq S} (-1)^{|S|-|T|},$$

which equals one when U=S and zero otherwise. Hence Φ_S consists exactly of monomials with support S. Such monomials have total degree at least $|S|$. If $H[S]$ splits into nonempty components, the restricted energy is a sum over those components, Z_S is a product, and its logarithm is a sum. It has no monomial meeting distinct components, so the exact-support part vanishes. Finally, the restricted exponents involve only edges inside S and the labels w_v. Partitioning all monomials by support proves (4), and specializing every $y_v = x$ proves the assertion. □

This is a sufficient-statistics theorem, not an inverse equivalence: equal decorated small-subgraph counts imply equal truncated series. Equal truncated series need not determine all those counts individually, since their polynomial contributions can be dependent.

Proposition 2 (exact recovery and fixed-budget cost). Write $F_\lambda = \sum_{j \geq 0} a_j x^j$ and $B_r = r[x^r] \log F_\lambda$. Then

$$a_0 = 1, \qquad j a_j = \sum_{r=1}^{j} B_r a_{j-r}, \qquad B_r \in \mathbb{Z}[q]. \tag{5}$$

For fixed R, maximum degree Δ and λ, an exact implementation of (3)–(5) has cost linear in n in an arithmetic-operation model. Its dependence on R and degree can be exponential; this is not a large-budget polynomial-time claim or a bound on coefficient bit cost.

Proof. The identity $xF' = F\, x(\log F)'$ gives (5). The inverse of a series in $1 + x\mathbb{Z}[q][[x]]$ has integer-polynomial coefficients, so xF'/F also does, proving the assertion for B_r. A connected s-vertex set has a spanning tree and a walk of length $2(s-1)$ visiting all its vertices. Choosing a root and such a walk bounds the number of sets by $n \max(1, \Delta)^{2(s-1)}$. For each set of size at most R, evaluating all restrictions in (3) takes at most 2^R restricted calculations. A restriction on t vertices contains at most $\binom{R+t}{t} \leq 4^R$ profiles of total height at most R. Energy exponents are bounded by $(\Delta+\lambda)R^2$. Truncated polynomial arithmetic, logarithms and (5) have cost polynomial in this exponent range and R per elementary profile step. One may therefore use the conservative bound

$$n\,[8 \max(1, \Delta)^2]^R \operatorname{poly}(R, \Delta, \lambda) \tag{6}$$

in coefficient operations. This bound requires no canonical labeling or graph-isomorphism oracle: omitting memoization preserves it. Exact memoization changes constants and sharing, not correctness. □

The implementation uses tree encodings for common decorated trees, exact small canonical forms when inexpensive, and a vertex-set key when canonicalization would be costly. The fallback loses reuse only. It never identifies different decorated shapes. This matters because factorial canonicalization can cost more than the profile arithmetic it is intended to save.

3 The first cycle visible to a budget

For a regular graph of degree d, put

$$a = q^{d+\lambda}, \qquad f = q^{-2} - 1, \qquad \kappa_{d,\lambda} = af = q^{d+\lambda-2}(1 - q^2). \tag{7}$$

The temporary Laurent notation f is harmless: each final expression below is a polynomial when d is at least two. The girth of an acyclic graph is understood to be infinite. Let $N_g(H)$ count unoriented simple g-cycles.

Theorem 3 (first-cycle coefficient; SNG-F56). Let g be at least three. Suppose H and H′ are d-regular graphs of the same order, both of girth at least g. Then

$$F_\lambda(H) \equiv F_\lambda(H') \pmod{x^g}, \qquad [x^g](F_\lambda(H) - F_\lambda(H')) = (N_g(H) - N_g(H'))\kappa_{d,\lambda}^g. \tag{8}$$

Proof. We first justify the tree counts used in the locality formula. For a tree T on k vertices, the number of all graph homomorphisms into a d-regular n-vertex graph is nd^{k-1}, obtained by choosing the root image and then one of d neighbors for each child. Group these maps by their equality partition of the vertices. After identifying equal vertices and suppressing parallel edges, the quotient is connected. A quotient with a loop has no map into a simple graph. For k at most g, any nontrivial quotient has fewer than g vertices; an injective realization of a cyclic quotient would violate the girth condition. Every contributing proper quotient is therefore a smaller tree. Induction on k now shows that the number of injective homomorphisms of T depends only on n,d and T for k at most g.

For k below g an injective tree image has no extra edge among its vertices: adding one to the tree would create a cycle on at most k vertices. Thus the number of induced copies of every such T is its injective homomorphism count divided by $|\operatorname{Aut} T|$, independently of H. All ambient-degree labels also agree. Theorem 1 proves equality of logarithms, and hence series, below degree g.

At degree g, contributions from supports smaller than g still agree. An induced graph on exactly g vertices is either a tree or a chordless g-cycle whenever it is connected. An injective tree image can have an extra edge only if its tree is the path on g vertices: the cycle formed by that edge must use every vertex. Each g-cycle has exactly g spanning paths. Consequently the induced-path count is the universal injective path-image count minus $gN_g(H)$; every other g-vertex tree count is universal.

Only height one at every support vertex contributes at total degree g. The coefficient of $\prod_{v\in S} y_v$ in $\log F$ is

$$a^g \sum_{\substack{D\subseteq E(H[S]) \\ (S,D) \text{ connected}}} f^{|D|}. \tag{9}$$

To verify (9), expand the binary weight on S as $a^{|S|}\prod_{e\in E(H[S])}(1+f)$. Partition each chosen edge set into its connected components. The exponential formula, or direct expansion of the formal logarithm followed by this component partition, leaves precisely connected spanning edge sets. A path contributes $a^g f^{g-1}$. A cycle contributes $a^g(gf^{g-1} + f^g)$, from its g spanning paths and its full edge set. Subtracting the g missing induced paths leaves $a^g f^g$ per cycle. This proves the logarithmic discrepancy. Since all lower coefficients agree and the constant coefficients are one, the first discrepancy of the series equals that of their logarithms. Equation (8) follows. □

For the cube and eight-vertex Möbius ladder, d=3, g=4 and the square counts are six and four. Equation (8) gives $2q^4(1-q^2)^4$ for L and $2q^{12}(1-q^2)^4$ for A. The former

is exactly the retained reference discrepancy. The triangular prism and $K_{3,3}$ give the analogous volume-three difference from their two versus zero triangles. These applications explain particular fixtures; they do not assert that cycle counts determine the entire height series.

4 Arbitrarily late collisions and closing contexts

Corollary 4 (cycle collision family). For every integer $g \geq 3$,

$$A_{C_g \sqcup C_g} - A_{C_{2g}} = 2\kappa^g x^g + O(x^{g+1}), \qquad \kappa = q^2(1-q^2). \tag{10}$$

In particular, the series agree below volume g and differ at g for every $q \in (0,1)$.

Proof. Both graphs have order 2g, degree two and girth at least g; their g-cycle counts are two and zero. Apply Theorem 3 with $\lambda = 2$. □

There is an independent transfer proof, useful for checking the leading term. At a fixed cutoff R, restrict indices a,b to $0, \dots, R$ in

$$T_{ab} = x^b q^{(a-b)^2+2b^2}.$$

A cyclic matrix product multiplies exactly one x-factor and one pinning factor for each vertex, so $A_{C_m} = \operatorname{tr}(T^m) \bmod x^{R+1}$. The finite-dimensional trace identity gives

$$(\operatorname{tr} T^g)^2 - \operatorname{tr} T^{2g} = 2\operatorname{tr}((\wedge^2 T)^g). \tag{11}$$

The column indexed by c<d in the exterior-square matrix carries x^{c+d}. Minimum degree one occurs only at $(0,1)$, whose diagonal entry is

$$x \det \begin{pmatrix} 1 & q^3 \\ q & q^2 \end{pmatrix} = xq^2(1-q^2).$$

Every closed product of g entries has degree at least g; equality forces every intermediate pair to be $(0,1)$. Thus (11) gives exactly the leading term (10). All matrices were finite, and enlarging R stabilizes every coefficient under discussion.

To state the context consequence precisely, form B_H by adding distinct nonadjacent terminals s,t, each joined to every vertex of H. A closing context C is a fixed finite simple graph with two designated ordered terminals. It is glued to B_H by identifying corresponding terminals and retaining all other vertices separately. Terminal edge conventions are unchanged by H, and the context has no additional identification with the interior. Write $Z_{C,H}$ for the resulting unpinned local height series.

Proposition 5 (context-independent first discrepancy; SNG-F53–SNG-F54 specialized). For interiors of the same order m,

$$Z_{C,H}(x,q) = \sum_{j \geq 0} A_{H,j}(q) x^j W_{C,j}(x,q), \qquad W_{C,j}(0,q) = 1. \tag{12}$$

Hence a first difference at interior volume g remains the same coefficient at volume g under every such closing context. In particular the family (10) cannot be distinguished earlier by any fixed context.

Proof. Assign heights s,t to the two terminals. For an interior profile of volume j, the edges from terminals to H contribute

$$m(s^2+t^2) + 2\sum_{v \in H} h_v^2 - 2j(s+t).$$

The first and last terms depend only on m,j,s,t, and the middle term is precisely the pinning in $A_{H,j}$. Summing terminal and context-interior heights defines $W_{C,j}$; it is a formal series with coefficients in $\mathbb{Q}(q)$. Its constant term comes uniquely from zero heights on both terminals and all context-interior vertices, and equals one. Thus (12) is a unit lower-triangular transformation of the A-sequence, common to all interiors of order m. It preserves the first nonzero difference. □

For g=6, the two dominating blocks have fourteen vertices. Closing each with a single shared neighbor gives fifteen-vertex graphs. Direct weak-composition enumeration of these actual glued graphs gives equality through volume five and, at volume six and $q = 1/2$,

$$2(\tfrac{1}{4} \cdot \tfrac{3}{4})^6 = \frac{729}{8\,388\,608}.$$

This is a finite independent check of the structural identity, not evidence that the complete series are equal: (10) proves they are unequal.

5 Sharp reconstruction of every 2-regular interior

For k at least one, let φ_k be the exact-support contribution (3) for a k-vertex path, with every vertex assigned ambient degree two and pinning two. Theorem 1 gives $\varphi_k = O(x^k)$, so

$$b(x,q) = \sum_{k\geq 1} \varphi_k(x,q), \qquad D_\ell = \log A_{C_\ell} - \ell b \tag{13}$$

are well-defined formal series, computable to every finite budget.

Theorem 6 (sharp 2-regular reconstruction; SNG-F57). Fix an order n and a real $q_0 \in (0,1)$. The coefficients of $A_H(x,q_0)$ through degree $R = \lfloor n/2 \rfloor$ determine the isomorphism type of every simple 2-regular graph H of order n. For every n at least six this uniform budget is sharp. Over $\mathbb{Q}(q)$, or after the same specialization, the cycle counts are recovered recursively by

$$\log A_H - nb = \sum_{\ell\geq 3} c_\ell D_\ell, \qquad c_r = \kappa^{-r}[x^r]\left(\log A_H - nb - \sum_{3\leq \ell<r} c_\ell D_\ell\right). \tag{14}$$

Proof. A simple 2-regular graph is a disjoint union of cycles of lengths at least three. For a cycle of length ℓ, each connected proper induced subset is a path. There are ℓ copies of each path size $1 \leq k < \ell$. Therefore Theorem 1 gives

$$\log A_{C_\ell} = \ell \sum_{k<\ell} \varphi_k + \Phi_{C_\ell}, \quad D_\ell = \Phi_{C_\ell} - \ell \sum_{k\geq \ell} \varphi_k.$$

In particular $D_\ell = O(x^\ell)$. The multilinear path contribution at size ℓ is $q^{4\ell} f^{\ell-1}$; the cycle contribution is $q^{4\ell}(\ell f^{\ell-1} + f^\ell)$. Their difference shows

$$D_\ell = \kappa^\ell x^\ell + O(x^{\ell+1}). \tag{15}$$

The series A multiplies over connected components, and logarithms add. Subtracting the total vertex background nb proves the first identity in (14). Equation (15) makes that identity triangular, proving its recursive inverse. At $q_0 \in (0,1)$, $\kappa(q_0) \neq 0$, so every division is valid.

Recover all cycle counts with length at most R. Two remaining cycles would each have length greater than R, requiring at least $2R + 2 > n$ vertices. Hence there is at most one remaining cycle, and its length is the remaining vertex count. The recovered multiset of lengths determines H up to isomorphism.

For sharpness, put $g = \lfloor n/2 \rfloor$. If n=2g, compare C_{2g} and $C_g \sqcup C_g$. If n=2g+1, compare C_{2g+1} and $C_g \sqcup C_{g+1}$. For n at least six all component lengths are admissible. Their logarithmic differences are respectively $2D_g - D_{2g}$ and $D_g + D_{g+1} - D_{2g+1}$, whose first nonzero coefficients occur at g by (15). Thus no smaller budget distinguishes every pair. For n=3,4,5 there is only one simple 2-regular isomorphism type, so no analogous lower bound is claimed. □

The hypothesis that n is known is part of the theorem. It is natural for fixed-order graph comparison and avoids hiding order recovery inside the final remainder step. The rational implementation takes exact input coefficients, validates the recovered nonnegative integer multiplicities and remaining length, and can reject data that are inconsistent with a 2-regular graph.

6 Reproduction and measured cost

The package uses Python 3.10 or later and only the standard library for its exact core. From the repository root, `PYTHONPATH=src python scripts/locality_family.py --check` reproduces the retained structural receipt in `results/locality_family.json`. The receipt compares the compiler with an independently implemented weak-composition enumerator on ten explicit or seeded graphs for both $\lambda = 0$ and 2, checks (8) against the prism and cube examples, verifies (10) for $3 \le g \le 8$ with independent enumeration through g=6, enumerates the fifteen-vertex glued example directly, and reconstructs all 57 cycle partitions of total order at most fourteen. The sharpness pairs for every order six through fourteen are retained too.

The manuscript's theorems are proved above in ordinary mathematics. The connected-support code and these general theorems are not formalized in Lean. Separate Lean artifacts for the first paper's fixed reference coefficients do not enlarge the formal-verification scope of this paper. Equality to an independent enumeration oracle is finite evidence about the implementation, not a machine-checked proof of its general correctness.

The following timings are retained wall-clock measurements, not portable performance guarantees. Graphs are connected cubic graphs produced by the seeded configuration-model generator in `scripts/locality_family.py`; n is the seed. The separator baseline uses the same total-volume and height cutoffs, a stated min-degree elimination order, and exact coefficient comparison wherever it is run.

n	Connected supports, R=4	Separator engine, R=4	Elimination width	Connected supports, R=6
16	0.027 s	0.243 s	4	0.900 s
32	0.025 s	52.611 s	7	0.818 s
64	0.037 s	not run	13	0.901 s
128	0.042 s	not run	27	0.796 s

At n=64 and 128 the R=4 baseline would allocate, over all bags under its current representation, approximately 8.19×10^9 and 4.66×10^{19} height assignments. These are

representation-size estimates, not measured runtimes or lower bounds for every tree-decomposition algorithm. The R=6 figures have still larger estimates and were not run. Observed reuse depends on how many decorated types occur; at R=6 these four graphs contain 32,26,29 and 17 types. This explains why the finite timings need not increase monotonically with n. The theorem guarantees linear graph-size dependence at fixed parameters even without reuse, rather than a sublinear guarantee.

To rerun timings use `PYTHONPATH=src python scripts/locality_family.py --benchmark`; this updates the retained timing section and is deliberately separate from deterministic `--check`. For the accompanying reference coefficients use `PYTHONPATH=src python scripts/reproduce.py --check`. The full archive includes sources, tests, exact JSON receipts, the bibliography and rendered manuscripts. The detailed claim register identifies every finite boundary and the editorial release check.

7 Scope and next questions

Finite agreement below volume g is now explained and attained by an explicit family, and within the cycle class a sharp inverse theorem is available. It does not follow that the fixed-pinning series reconstructs every finite graph. Nonisomorphic graphs with equal complete A-series remain a legitimate target. The connected-support identity supplies a sufficient condition for truncated equality, but its aggregated q-polynomials can merge different decorated statistics. An exhaustive prefix search therefore cannot be promoted to a completeness theorem.

Past the first cycle, equal cycle counts need not imply equal coefficients: overlapping cycles and other connected footprints contribute. The project has additional loop and overlap results (SNG-F60–SNG-F61) and a complete moment enrichment (SNG-F58). They are deliberately outside this paper's central proof and are not prerequisites for Theorem 6. A further algorithmic question is to combine separators inside the connected footprints with the fixed-budget compiler, extending the useful budget without requiring bounded treewidth of the full graph. These questions retain the same explicit volume, parameter and graph-class boundaries as the proved results.

References

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